

Development of Electromyography-Controlled Assistive Robotics for Self-Feeding

Team I: AutoBite

Authors: Alexandra Jensen, Anna Moulds, Jacob Pearl, Zaid Slayyeh, Julia Stoneburner, Joshua Yap

Grader Note: Our paper structure is based on design and evaluation for *IEEE Transactions on Neural Systems and Rehabilitation Engineering*. Similar papers can be found here:

<https://www.sciencedirect.com/science/article/pii/S0921889018307061>

<https://journals.sagepub.com/doi/full/10.1177/17298806231183571>

Abstract

Our project aims to develop an assistive feeding device that will restore independence to individuals who cannot self-feed due to a disability. Eating is an essential part of daily life, and the inability to eat for oneself can negatively impact autonomy and well-being. Even with advancements in assistive technology, meal assistance remains underexplored, as existing solutions primarily rely on serial linkage robots with safety concerns and lack integration with other caregiver tasks. Our initiative, Auto-Bite, aims to close this gap by assisting with feeding using a mobile manipulator robot. Stretch is a mobile manipulator developed by Hello Robot that interacts safely with humans. Our project is creating a user-friendly and safe solution for assistive feeding, combining the safety of the Stretch robot with the adaptability of electromyography sensors and an attached graphic user interface for customization. We developed a system of hardware and software to process EMG signal data and communicate it to the Stretch robot. Along the way, EMG signal data was collected and tested with several digital filters to prevent false activations and tailor signal strength to user capability.

Additionally, pre-programmed trajectories were implemented to perform smooth and repeatable actions for the feeding motion, and a prototype of a graphic user interface was developed. Ultimately, using EMG proved to be a safe and adaptable solution for controlling assistive feeding robotics, and our Stretch robot successfully follows a repeatable feeding trajectory. This promising model can be further developed to improve the quality of life for users of assistive devices and their caretakers.

Introduction

Background

Disability is widely understood through the medical model of disability, which conceptualizes disabilities as intrinsic limitations that require correction through medical or technological treatment. In contrast, an alternate mode of thinking is the social model of disability, which argues that disability arises from the interaction between individuals with impairments and a society that fails to provide adequate accommodations for their needs (Williamson, 2015). Under this framework, impairments are defined as attributes or functional limitations, such as paralysis or the inability to walk, while disabilities represent the restrictions experienced when the needs of people with impairments are unmet. If our society were inherently designed to be accessible, individuals with impairments would not experience

disability (Ralph, 2015). In developing our robotic system, we aim to follow the social model of disability by bolstering society's accommodations for people with disabilities.

Building on the principles of the social model of disability, one critical area needing societal accommodations is the ability to eat. Authors of *The Anthropology of Food and Eating* state that, "Next to breathing, eating is perhaps one of the most essential of all human activities, and one with which much of social life is intertwined" (Mintz and DuBois, 2002). As eating is a crucial activity in our lives, the inability to feed oneself has been shown to lead to feelings of incompetence and shame (Hermann et al., 2019). A qualitative study done on the eating experiences of people with disabilities reported many negative emotions and outcomes as a result of being fed or struggling to feed oneself. Benito, 71, with acquired brain damage, stated: "They told me off all the time—if I ate too slowly, if I did not eat, and eventually they fed me themselves. They made me eat too fast, and the food did not sit well." Paloma, 60, with amyotrophic lateral sclerosis, recalled, "While I eat, I drop a lot of food, and once I heard a man saying aloud that it was disgusting, and he turned his back on me so he would not see me. Since that day, I prefer to eat when I am alone." Many participants reported feelings of loss of social settings and pleasures surrounding meals, as well as strong feelings of being a burden. As a result, these feelings led to lower self-esteem and social withdrawal as public meals were avoided (Cipriano-Crespo, 2020) (Lance, 2007).

Various conditions affecting upper limb mobility can contribute to the physical inefficacy to feed oneself, such as spinal cord injury, cerebral palsy, amyotrophic lateral sclerosis, amputation, and arthritis. For spinal cord injury alone, approximately half of the 294,000 Americans living with the condition experience it at the cervical level, contributing to a loss of hand and upper limb function, which may impair feeding efficacy (Radioff, 2022). In the United States, approximately 1 in 4 people have a disability, and 1 in 8 people have a disability that affects their mobility (CDC, 2015). It is, therefore, essential to develop assistive feeding devices that can restore independence to individuals who cannot feed themselves due to limited upper limb mobility or otherwise reduce dependence on caregivers.

Caregiver availability is declining due to aging demographics, which increases those in need of care (Naotunna, 2015), in addition to high turnover rates, resulting in a widening gap in hands-on care (Smith et al., 2021). Furthermore, the cost of caregiving is high, both financially and physically. Approximately 26% of caregivers spend their personal income on caregiving endeavors, and 12% take out a loan or borrow money from friends and relatives (Goad, 2025). An AARP report states that 23% of Americans say that caregiving worsened their own health and that 61% of family caregivers are also working (AARP, 2020). There are approximately 39.8 million caregivers in the United States who provide unpaid care to adults with disabilities or illnesses (Coughlin, 2010). These informal caregiving services were estimated to be worth \$470 billion in 2013, exceeding the total Medicaid spending for that year (Reinhard et al., 2015). These statistics indicate that there is an increased demand for assistive technologies, such as feeding robots, that can supplement or replace human caregivers in delivering critical daily care (Jones & Miller, 2023) and improve the autonomy and quality of life for people with disabilities that affect eating.

Gaps in Knowledge & Motivation

The research and development of robotics for assistive feeding and meal assistance still face limitations in effectiveness and usability. Most studies focus on serial linkage or "articulated" robotic arms with joints that allow for rotational motion. The popularity of this type of robotic arm is due to its high degrees of freedom (DOF) that the rotating joints permit, allowing for it to be manipulated in many different ways and enabling a diverse range of applications and functionality when it comes to life tasks. However, the high power produced by serial linkage robots is better suited for the industrial applications they were originally produced for and poses concerns for human safety for in-home use (Matthias et al, 2011).

For our project, we will explore the use of a mobile manipulator as an alternative type of robot for assistive feeding. A mobile manipulator robot can move linearly along X, Y, and Z coordinates, may have a rotation at the 'wrist,' and can maneuver around its environment on a wheeled base. This is unlike a serial linkage robot, which moves by rotating along joints between arm segments. Although the mobile manipulator robots have fewer degrees of freedom than those of serial linkage robots, their mobile base enables a range of access that is sufficient for in-home tasks. Additionally, mobile manipulator robots are specifically designed to be able to operate alongside humans safely using built-in advanced sensors and machine vision to avoid obstacles or harmful contact with the user. Features like these reduce the risk of accidents, which can lead to more opportunities for combined human-robot achievement or human-in-the-loop technology. Therefore, it is a safer alternative when compared to traditional robots that require isolated workspaces.

In our paper, we decided to develop the mobile manipulator Stretch 3 (**Figure 1**), developed by Hello Robot. The robot is a mobile manipulator that is already on the market and used in homes as an assistive device. Several key features influenced our selection. The Stretch is very user-friendly as it has a compact footprint to navigate clutter and tight spaces, is fairly lightweight, and has contact sensitivity to allow for safe operation alongside humans. Furthermore, it is almost entirely open source, with a large user community that shares items such as code, simulations, and research papers for the mutual benefit of all users. Finally, it is highly dexterous with advanced teleoperation capabilities, along with an advanced wrist with three degrees of freedom and a compliant gripper. All of these features enable the Stretch to maneuver around the home and complete daily tasks such as locking the door, picking items off the floor, delivering food or medication, and more (Ackerman, 2023).



Figure 1. *The Stretch* by Hello Robot

One task that has yet to be fully developed for this particular robot is meal assistance, in which the robot brings bites of food to the user's mouth. The Stretch is a device that has many functional tasks already, and adding a feature for feeding assistance is one that has had limited development and coverage. Previous studies developed by the creator of the Stretch implement the task of feeding but are not widely available for public use. (Olatunji et al., 2023). Still, there

needs to be documentation or sources for how the process of coding the robot was done, and thus is not fully developed. The feeding task that already exists within the Stretch uses manual input via directional controls. Most commercial assistive feeding robots use a keyboard and mouse or joystick user control system and have the shortest learning curves (Naotunna et al, 2015). However, manual controls are not always intuitive, as the robot has more degrees of freedom of movement than the human operator can readily control (Losey et al., 2020). Automating the feeding motion while the timing of each bite remains in the control of the user can, therefore increase the usability of the robot while maintaining autonomy.

System Overview

In this paper, we introduce an autonomous feeding system that enables a general-purpose mobile manipulator, the Stretch 3 robot, to provide safe, easy-to-use meal assistance. The system facilitates autonomous feeding assistance, where the Stretch 3 follows pre-programmed trajectories to move food from a plate to a user's mouth while maintaining safety and efficiency. The system is designed to accommodate individuals with limited upper limb mobility by offering smooth and repeatable feeding motions through trajectory planning and EMG sensor-based activation. As the device is intended for assistance with feeding, it should not be used unassisted for users who also require assistance for chewing and swallowing.

The feeding process consists of three primary subtasks: food acquisition, where the robot moves toward the food and may execute a stabbing motion for secure pickup; food delivery, where the robot transports the food to the user's mouth via a predefined trajectory; and return to the initial position, preparing for the next feeding cycle. The EMG sensor input serves as a control mechanism for initiating these motions, ensuring accessibility for users with varying levels of mobility.

We designed an accessible two-part input system with an electromyography muscle activation sensor to command the robot during feeding, alongside a graphical user interface (GUI) to ensure the robotic system can be tailored to individual needs and preferences and allow for even more autonomy. Various muscles can be leveraged to operate technology through a suitable system. The EMG sensors detect muscle activation to serve as a "go" command, initiating the feeding interval by signaling the Stretch robot to approach the user's mouth. These sensors can be placed on various muscles based on the user's preference or physical capability, enabling effortless detection without requiring strenuous movement. EMG does not require that the user have full ability to move a muscle but can operate from flexing the muscle alone, which broadens the user base (Yamagami et al., 2023). Secondly, a graphical user interface (GUI) is developed to be used before or after feeding to make adjustments to the user's feeding preferences. Through the GUI, an individual who is using the system can tailor the threshold sensor read value. This allows users with motor impairments or their caregivers to adjust feeding parameters, including movement speed and muscle activation thresholds.

To ensure safety and reliability, our system incorporates trajectory planning constraints, real-time EMG signal filtering, and emergency stop mechanisms triggered by user input. By prioritizing safe, compliant arm motions and integrating multimodal execution monitoring, the system proactively prevents potential unexpected executions during the feeding process. It should be noted that some degree of muscle activation is needed to use our stretch 3 program, so an individual with conditions such as paralysis or similar, would not be the targeted demographic for the Auto-Bite project.

Materials and Methods

Stretch Body Experimental Setup

The robot is positioned at a table across from the 'user,' a mannequin head that serves as a human proxy (**Figure 2**). The Stretch holds a fork in its grippers, and a food item is placed in front of the robot, and the user. At a start command, the robot executes a feeding motion of picking up the food and moving it toward the mannequin's mouth before returning to the starting position.



Figure 2. The experimental setup features the Stretch holding a utensil, a table and bowl, a mannequin head at the approximate height of a user, and a monitor printing the robot's joint angles.

The Stretch was intentionally placed across the table from the user as this allows for the robot to move within a single plane. This positioning is beneficial as it reduces the degrees of freedom that the robot would need to utilize otherwise. The linearity of the arm makes it such that reaching across the table is the simplest and least extraneous motion. If it were instead beside the user, the base of the robot would have to be the primary motion for feeding which moves the whole robot, and this could be frustrating for the user and those surrounding in a dining setting. Opting for the least amount of motion is also inherently safer.

Measurements of the experimental setup were done so that we could create a trajectory for the feeding motion. Measurements included the height of the mouth, the distance across the table to the mouth, the position of the bowl, and several others. Within the script for the feeding trajectory, these measurements were input in meters to create a trajectory with an initial position of the utensil held vertically at the bowl and a final position of the utensil held horizontally at the user's mouth.

The robot follows a predefined point-to-point trajectory to assist with feeding during meals. The meal assistance cycle consists of the following steps shown in the workflow in **Figure 3**:

Food Acquisition: The robot moves its end effector toward the food. A "stabbing" motion may be included to pick up solid food items.

Food Delivery: The end effector follows a trajectory from point A (food location) to point B (mannequin mouth) while maintaining a safe and controlled speed.

Return to Initial Position: After reaching point B, the robot retracts back to point A after input to prepare for the next feeding cycle.

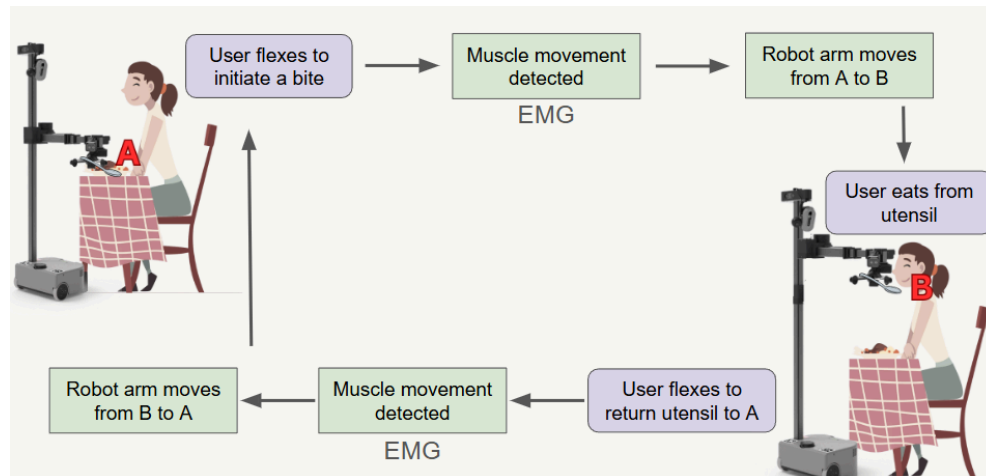


Figure 3. Basic workflow of the robot-assisted feeding cycle.

Each action is performed based on the user commands as detected by EMG. If the user contracts the measured muscle to send a signal, the robot will perform the appropriate task depending on its current state. The EMG thus acts as a command signal for certain tasks. For instance, the user may flex one muscle to initiate feeding but may flex another muscle to stop or even return the robot to the plate.

EMG System Components

To acquire EMG signal data, a proof-of-concept EMG sensor setup was established. To ensure consistency across all trials during the data collection period, a full setup protocol was established, including cleaning the skin, EMG electrode placement, and amplifier setup (Appendix I). The setup protocol was informed by *The ABC of EMG* by Peter Konrad (2005).

Electrodes and placement

Pre-gelled electrodes were chosen to reduce setup time and the potential for setup error. To collect initial data, we chose to collect data from two muscles simultaneously to test co-contraction. Wrist extensor and flexor muscles were chosen for user ease of access. One pair of electrodes is placed on the extensor muscles, the "top" side of the forearm, with the centers 2 cm apart. Another pair is placed on the flexor muscle, the "bottom" of the forearm, also with the centers 2 cm apart. The ground electrode for each pair is placed on the bony area (no muscle) of the elbow (**Figure 4**).



Figure 4. EMG sensor setup on forearm flexor carpi radialis muscle.

Wiring and Amplification

The two groups of electrode wires are connected to the Biopac IPS100C Isolated Power Supply. Signals are coupled to the HLT100C module and then are connected to an Arduino UNO's analog pins using the OUTISO cables (**Figure 5**). Arduino UNO was implemented as it is widely available and simple to use, and our team was most familiar with programming it and collecting data with its interface out of other commercially available microcontrollers.

From the Arduino UNO, the signals are then sent to a computer, which is read by the Arduino IDE serial port reader script and graphed using CoolTerm, an open-source plotting and data collection software. Before applying filters to real-time data, we collected sample numerical data and tested the effects of several digital filters to find the best one for our application.

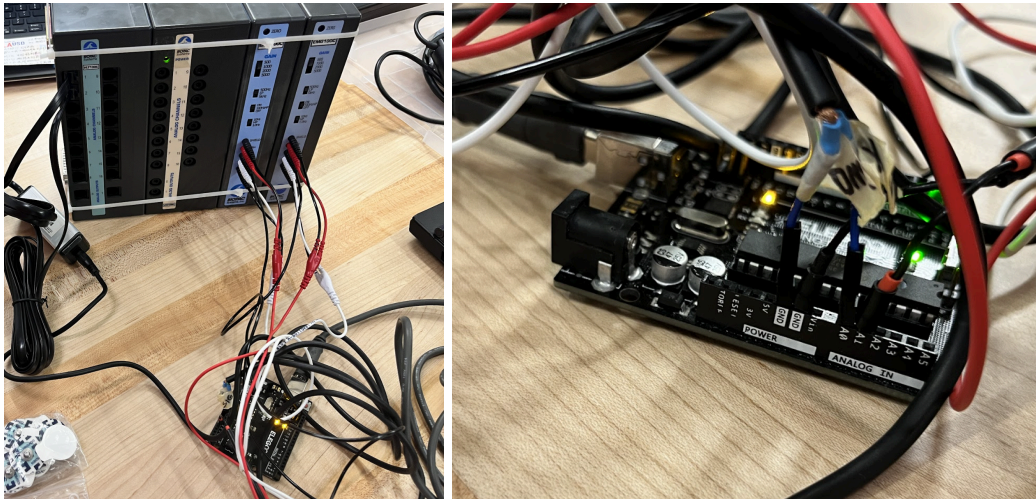


Figure 5. Biopac IPS100C and HLT100C modules connected to Arduino UNO analog pins.

Signal Processing

Rectification

The first step in EMG signal processing is rectification, which involves converting all negative amplitudes into positive values. This is achieved by reflecting the negative spikes symmetrically around the baseline, which is positioned at zero. As a result, the entire signal remains in the positive domain, eliminating the issue of fluctuating polarity (**Figure 6A**). Rectification not only improves the interpretability of the signal, but also allows for the application of standard amplitude-based parameters, such as the mean, peak (maximum) value, and total area under the curve, which are commonly used to quantify muscle activity. Rectification is also an essential preprocessing step for further signal analysis often in the form of smoothing and envelope detection. These processes involve applying digital filters to reduce high-frequency fluctuations and extract the overall trend of muscle activation over time to enhance the clarity and usability of EMG data for research applications.

Digital filters

Filtering is a critical step in EMG signal processing due to the signal's natural fluctuations and susceptibility to noise. Filters are used to smooth the signal, making the recorded data easier to interpret by reducing any rapid, unpredictable variations. Sharp amplitude spikes are smoothed out, allowing the overall trend of the signal to be highlighted. This is essential for accurately analyzing muscular activation patterns and ensuring reliable feature extraction.

There are numerous types of digital filters, each designed for specific processing needs. However, we selected three widely used filters to focus on due to their effectiveness and prevalence in the field. These are the moving average filter, the root mean square filter, and the Butterworth filter.

The moving average filter is a simple type of finite impulse response filter, meaning that it does not utilize feedback (no use of recursion). It smoothes a signal by taking a sliding window over the data and averaging a fixed number of past samples at each step. This number of samples is defined as the window size and can be adjusted based on the desired application. Its primary strengths are that it is simple and efficient while remaining highly effective for removing high-frequency noise and smoothing the signal. However, one potential drawback is that edges and sudden changes in the signal are smoothed out, which might be harmful in certain applications as detection/activation could be overlooked.

The root mean square filter is another type of finite impulse response filter, operating similarly to the moving average filter. It likewise uses a sliding window and averages a fixed number of past samples. However, the primary difference is that it squares the value of each data entry before summation, and then takes the square root of that sum. In doing so, unlike the moving average, this RMS filter maintains the overall energy level of the signal. This enables higher-amplitude components to be emphasized more, highlighting the larger variations. This can be useful for detecting muscle activation, which is usually characterized by an initial spike upon the first movement. However, some potential disadvantages are that it is more computationally expensive compared to a simple moving average filter, along with its greater sensitivity to outliers.

The Butterworth filter, unlike the previous two filters, is a recursive infinite impulse response filter. This means that the output at each time step depends not only on the current and past inputs, but also on past outputs. This differs from the finite impulse response filters, which do not account for any past outputs. It is designed to have a maximally flat frequency response, ensuring an accurate representation of low-frequency components. It functions by setting a desired cutoff frequency and then assigning various coefficients to adhere to that cutoff. It has a very strong and efficient smoothing response and also has a flexible design to be

adapted for different filtering needs. However, it is not ideal for high-precision applications, as it can allow some undesired signal components to pass through and can experience phase distortion.

We applied each of these filters to a sample sequence in which the user applied contractions of varying strengths to analyze which filter would be the most suitable for our EMG signal processing. To effectively utilize these filters, it is necessary to define key parameters that will influence their performance. For both the moving average and root mean square (RMS) filters, the primary adjustable parameter is the window frame size, which defines the number of past samples used for averaging. We experimented with various window sizes, ranging from twenty to fifty samples, to evaluate their impact on signal processing.

There are trade-offs between using a smaller versus a larger window size. A smaller window provides a more precise output, as it captures short-term variations with higher accuracy since the associated range of time is of shorter duration. However, a larger window incorporates more past data points, resulting in a stronger smoothing effect that reduces fluctuations. For our application, we decided that the larger window size (50 samples) was optimal, as it effectively minimizes signal fluctuations. If excessive fluctuations are present, it could cause unintended “deactivation” of the signal as it drops below the action threshold. Therefore, by using a smoother signal, we can ensure that the output will remain stable and stay above the threshold for the intended duration. Several example graphs displaying these window adjustments are shown in **Figure 6B and 6D**.

For the Butterworth filter, the primary adjustable parameter is the frequency cutoff. We utilized the Butterworth as a low pass filter, in which only the frequencies below the cutoff would pass through the filter, while the frequencies above the cutoff are eliminated. The goal is to define a cutoff frequency that effectively balances noise reduction with signal preservation. As EMG signals primarily contain useful frequency components under 500Hz, depending on the muscle type and movement, the most relevant muscle activity information can be found on a slightly truncated range. Frequencies above that range often consist of unwanted noise from sources such as electrical interference or artifacts but can be attenuated by the Butterworth filter. From testing with various frequency cutoffs, we found 400Hz to be the optimal cutoff frequency in striking a balance between signal clarity and noise reduction, retaining essential muscle activity information while filtering out unwanted high-frequency noise. A sample recording with this parameter is shown in **Figure 6C**.

The moving average filter exhibited the strongest smoothing effect, while the Butterworth filter exhibited the strongest preservation of peaks. We ultimately decided to utilize the root mean square filter since it still preserved the most major peaks while providing sufficient smoothing for our purposes.

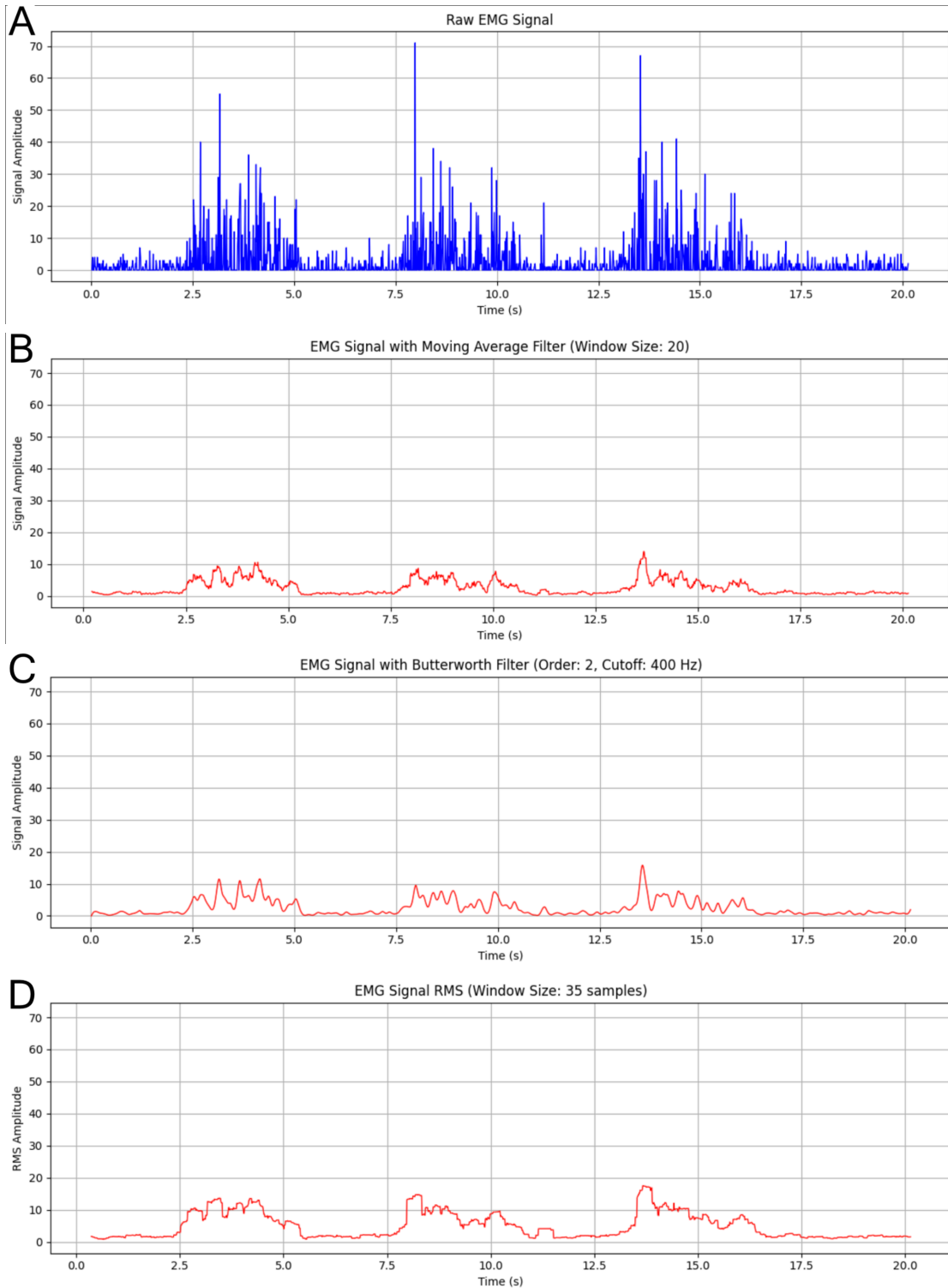


Figure 6. Comparing different filters on unfiltered EMG data. **A:** Unfiltered EMG reading from three periods of flexor carpi radialis activation (between neutral and regular, full-range wrist extension). **B:** Moving average filter applied to the data collected from Figure 6A, with a window size of 20. **C:** Butterworth filter applied to the data collected from Figure 6A, with an order of 2 and a cutoff frequency of 400Hz. **D:** Root mean squared (RMS) filter applied to the data collected from Figure 6A, with a window size of 35.

Writing to Data File, Visualization

To graph and record our data, we used an Arduino to process signals from the EMG. We initially graphed our data with Arduino IDE but ran into issues with the serial plotter and saving data to a file. We looked into different programs and methods others in the field had used. After some research, we came across a widely used open-source application called CoolTerm, developed by Roger Meier's Freeware, which outputs a much better plot for our data and allows us to write and stream our live data to a file, giving us our **Figure 6A**. It also provides the data file that we applied our filters to, to create **Figure 6B-6D**. We are using CoolTerm for an easy way of graphing our data and allowing us to explore different methods of filtering data. This will allow us to determine which method will best suit our needs for indications of muscle activations to trigger a pathway response from the robot. It should be noted that CoolTerm is to be used in our pre-application assessment and won't be used for the final Stretch 3 assistive feeding process.

User-specific parameters

Individuals who may want to use assistive feeding devices will have various comforts and familiarities with such technologies, as well as physical capabilities in their muscles. We have taken several steps to ensure our system is adaptable to user preferences.

We are developing an initialization sequence for the EMG sensors that will take place before feeding to prioritize user comfort and prevent false activations. This will involve first collecting data while the user is resting, which will display noise. Then, the user will flex with maximal voluntary contraction, optionally using a counteractive force, and measure the signal. The amplitude of the resting noise will be mapped to a value of zero, and the amplitude of the maximum contraction will be mapped to a value of one. In doing so, we create a range from 0-1 that represents all possible signal outputs from the user. After this mapping is established, the next step is to set a threshold frequency of activation. This value is derived from the signals of the resting noise, which are analyzed to find the magnitude of one standard deviation. The threshold is then set to be two times that magnitude, based on advice from our advisor Dr. Veronica Santos and under the guidelines of *The ABC of EMG* (Konrad, 2005). Since this threshold will be greater than the noise amplitude, any signal that is recorded to be greater than the threshold will be considered a muscle activation. This activation is subsequently recorded and transmitted to the Stretch Robot to trigger movement.

Our system can be further personalized depending on the muscle chosen for activation. The EMG electrodes can be placed on any muscle whose activation can be felt through the surface of the skin. Variability of the muscle chosen for activation monitoring by electrodes is crucial as there is a spectrum of physical limitations in users with disabilities. Flexing a forearm muscle might be comfortable for some, but shrugging a shoulder could be more viable for others. Therefore, we allowed a user to prioritize muscles they can activate the most comfortably.

These parameters will be unique to each user, as this initialization sequence can be used separately from the feeding function. In doing so, we allow the system to be customizable to each individual and inclusive of varying muscle activation capabilities. This threshold value will be part of the graphical user interface in development for users to implement customization before and after feeding. Other features of the GUI will include adjusting the feeding interval speed and creating meal "presets" where food textures and shapes can be entered. Individuals with limited upper limb mobility have various options available to control a computer mouse without using their hands, including head trackers, eye trackers, and sip and puff systems, so we assume that the user will have access to one of these or will get assistance from a care partner to make inputs in the GUI. As mentioned before, potential users of robotic feeding systems have varying levels of muscle strength in the body.

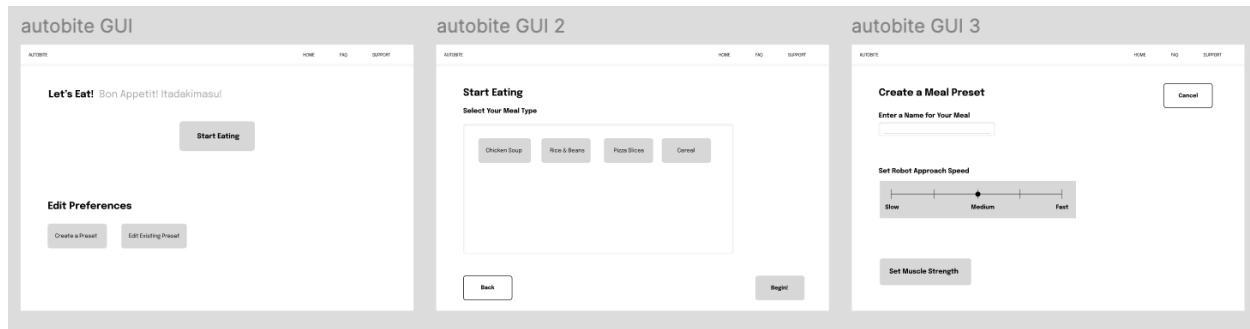


Figure 7. Mockup of a graphical user interface for customizing robot system preferences.

Data transmission to Stretch Robot

To communicate EMG signals to the Stretch robot in real-time, we established several integrated components. Once the preferred muscle is selected and the EMG is properly set up, the next step is to run the initialization sequence that we developed to set the trial-specific activation threshold. This is determined by recording noise as the user stays still, with the threshold being set above these recorded values. In **Figure 8**, this threshold is represented by the dotted green line. This setup is unique as it allows personalization to both the user's capability and environmental noise. Before each trial, this initialization sequence is prompted, allowing for the signal processing to be tailored to each individual's muscular signal noise. This enables our system to be fully inclusive, and able to be used by people with all different levels and locations of muscular ability.

After this threshold has been established and the initialization sequence concludes, the trial proceeds to the data acquisition phase. The raw EMG data can be seen in the light blue signal, demonstrated as a full contraction by the user. After we applied our selected filter, which was the root mean square filter, we obtained the dark blue signal, which we then used to determine when a contraction had occurred. The filter helps eliminate spikes in data, more accurately representing the intent of the user. When the user is not activating the muscle, an "off" state is communicated to the Stretch robot in the form of zeroes. Once the user flexes or activates the muscle for a full second, the "on" state is sent to the robot in the form of ones.

We also instituted a one-second time threshold, where the filtered signal has to be greater than the activation threshold to constitute a contraction. The purpose of this required contraction period is to bypass the occurrence of any accidental contraction of muscular spasm. A visual representation of this effect is present through the peak on the right side of **Figure 8**. The filtered signal is shown to surpass the activation threshold but does not trigger an "on" state. Alternately, when the filtered signal remains above the activation threshold for a full second, the "on" state is triggered, which sends a command to the Stretch robot to direct it to move.

CoolTerm was once again used to save and visualize this data, although it was not used during the real-time signal processing and data transmission. For the actual system, the Arduino Uno was connected directly to the base of the Stretch robot using a COM port and could transmit data directly to our robotic movement program.

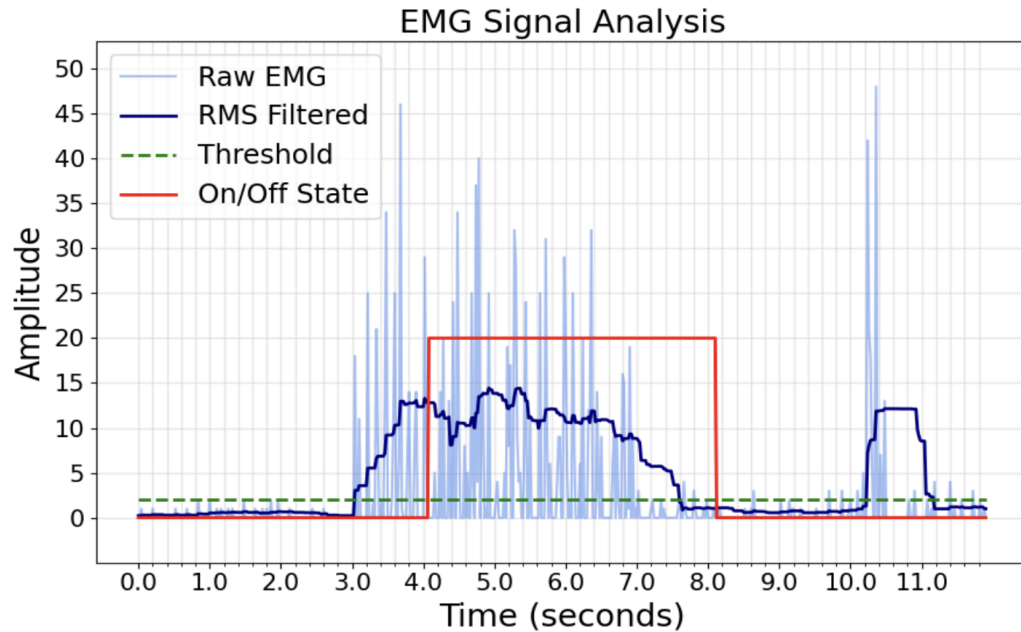


Figure 8. Real-time EMG Signal Processing and Transmission to Stretch Robot

Robot Hardware

Our robot-assisted feeding system hardware consists of the Stretch Robot 3 model and the three-degree-of-freedom dexterous wrist end effector. Given the adaptability and flexibility of the gripper on the end effector, a fork is placed in between the grippers manually before feeding. The robot is placed in a specific position given by the markers placed in the experimental setup, and the initialization and feeding functions begin through code execution.

Robot Software

Hello Robot's *stretch_body* library:

The code to control the Stretch robot's feeding movements was written using the *stretch_body* library, a Python-based Software Development Kit (SDK) developed by Hello Robot specifically for the Stretch. This software is designed to interact smoothly with the hardware of the Stretch Robot and comes pre-installed on the robot's Intel Nuc computer running Ubuntu 22.04, along with all the necessary libraries and software versions.

Stretch_mujoco simulation:

Due to the steep learning curve required to use the ROS simulation that Hello Robot supports for the simulation of their robots, we were restricted to using a less-supported Python-based simulator using the package, *stretch_mujoco*. Unfortunately, this physics-based simulator is under-supported and lacks key functionalities for our purposes. It was limited to simple *move_to* functions and could not implement scripts using trajectories or inverse kinematics.

To get around this issue, initial development began on our team members' personal computers using the *stretch_body* library within the *stretch_mujoco* environment through a GitHub repository. The *stretch_body* library was chosen because it included the necessary trajectory functions. This format was unsuccessful due to *stretch_body*'s need for specific configuration files that are only available on the Stretch robot's computer. One workaround would have been to download the files from the robot, but due to time constraints and the fact that *stretch_body* worked on the robot's computer, we abandoned the development of the

trajectories and movements in the *stretch_mujoco* environment. Instead, real-world development was furthered with the robot itself.

Trajectory Code:

Within the *stretch_body* library, there are several pre-made scripts for developers to run and modify. The file *stretch_trajectory_jog.py* provides a cubic spline function to give you trajectory coordinates for the robot to travel along a smooth path. As smooth, controlled movements are ideal for feeding actions, this script was modified for our purposes to fit the parameters of our experimental setup. The beginning and endpoints of each necessary joint position were input, with corresponding time points for when certain positions should be achieved. For instance, the robot's arm should extend to the user's mouth around the same time that the lift joint has reached the height of the user's mouth. Thus, the script's cubic spline trajectories were calculated, such as in **Figure 9**.

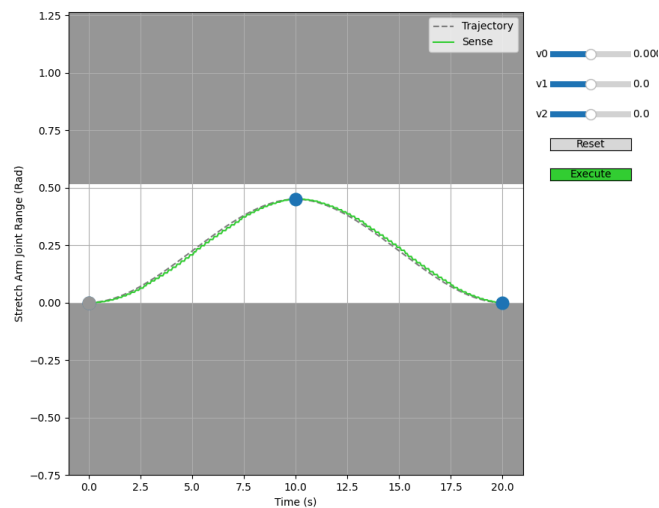


Figure 9. An example of a trajectory plot for the arm joint. Beginning, middle, endpoints, and velocities are given to create a cubic spline for the position with respect to time. Here, the arm moves out to 45cm in 10 seconds and returns to the initial zero position at the end of the 20 seconds.

Trajectories were input as such for the bowl-to-mouth, mouth-to-bowl, and food-stabbing motions. Given the position of each joint in the robot in time, a smooth trajectory was created at safe velocities that would not exceed the internal limits of the robot's servos. The cubic interpolation creates a continuous and smooth path that is predictable and slow, which are key features for keeping the robot program safe to use. The robot's trajectory can also be seen in **Figure 10**, which shows the position of the utensil through the food delivery command.

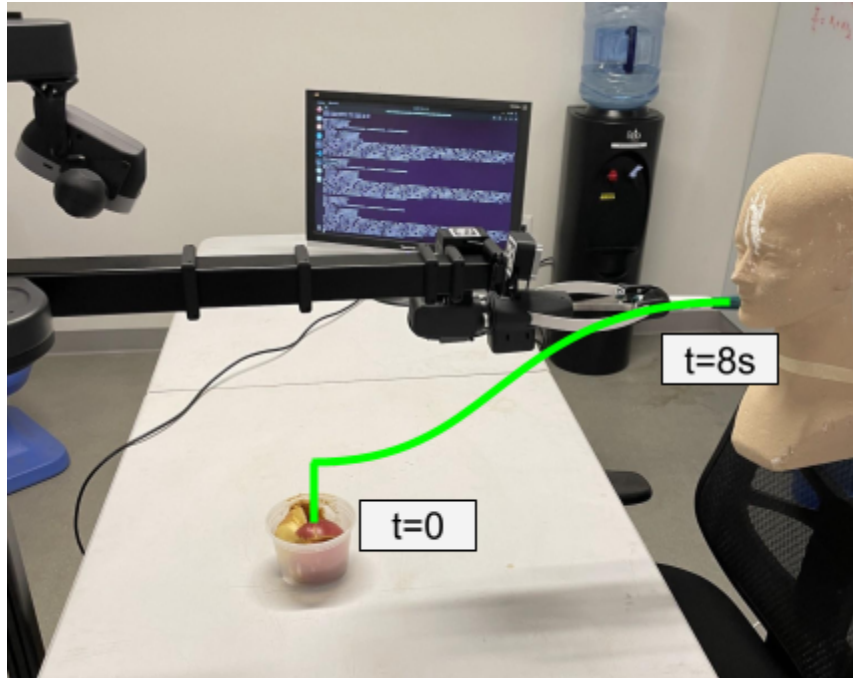


Figure 10. Robot's trajectory over time during food delivery.

Results

As a result of our development process, we successfully integrated EMG signals with predefined robot trajectories to create a proof-of-concept assistive feeding robotic system that functioned as intended.

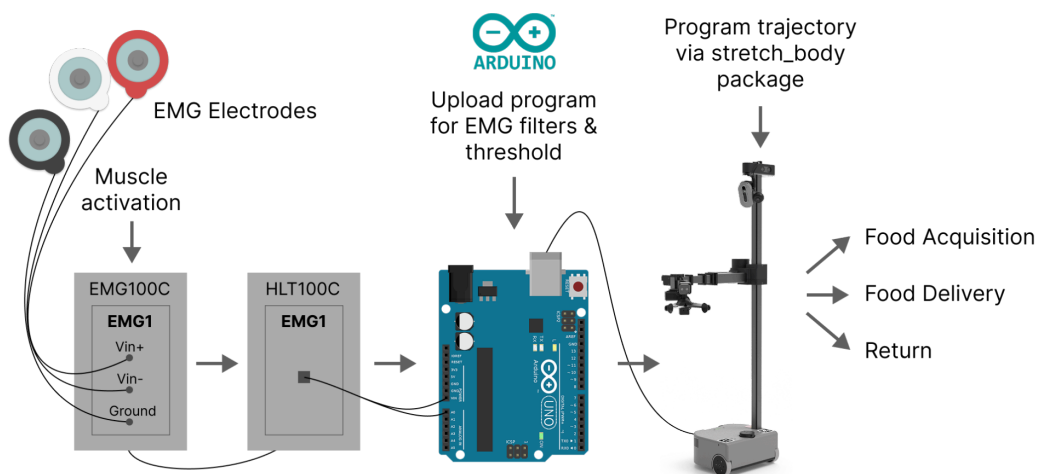


Figure 11. Schematic showing the communication of signals throughout the system.

Muscle signals are read through the electrodes, which can be placed on any surface muscle in the body that is comfortable to activate. These signals are then amplified through the BIOPAC EMG100C and HLT100C modules, digitally filtered through the Arduino UNO, and then communicated to the Stretch robot as an “on” or “off” state. The Stretch robot will proceed with the next step of the feeding process after receiving an “on” state when it is not actively moving in one of the feeding trajectories. The real-life setup of the assistive feeding system can be seen below, in **Figure 12**. Mirroring the schematic figure (**Figure 13**), it includes the EMG electrodes, BIOPAC modules, Arduino Uno, and the Stretch gripper. Our selected sample food for testing was marshmallows, due to their ease of being picked up by a fork.

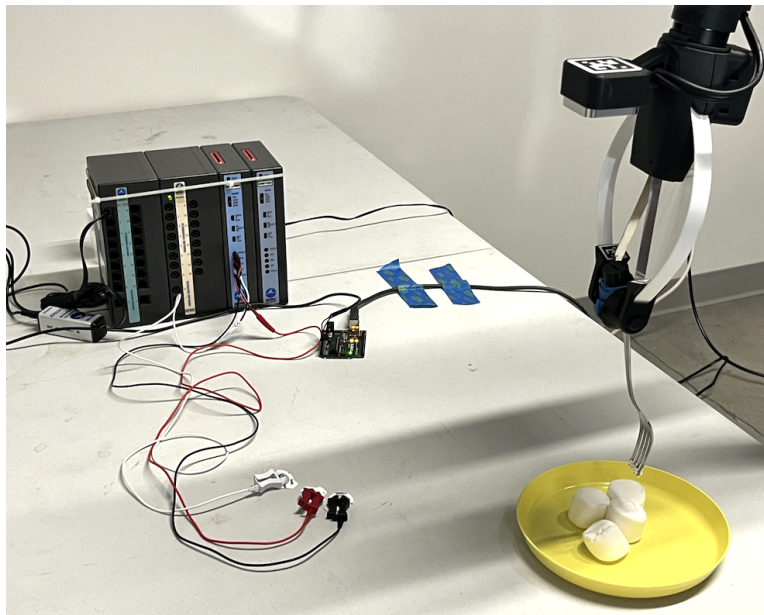


Figure 12. Final setup for integrated EMG sensors and Stretch 3 Robot

When beginning the feeding sequence, the user is prompted to “stay still” for three seconds as the EMG sensors collect noise (ideally with GUI but through the robot program for us). The program calculates an appropriate threshold value, unique to that trial, or meal. The system is now ready for activation.

If the user flexes the muscle for one full second above the threshold, the Arduino communicates an “on” state by sending an integer value “1” to the Stretch robot, which instructs it to complete the first step of the feeding process, sending the end effector downwards towards the food. While the robot is completing this step, it does not read any incoming signals. After food acquisition is completed, the user can flex again to have the robot deliver the food from the plate to the user's mouth. When they have eaten off of the utensil, they may flex again to send the robot back to its initial position.

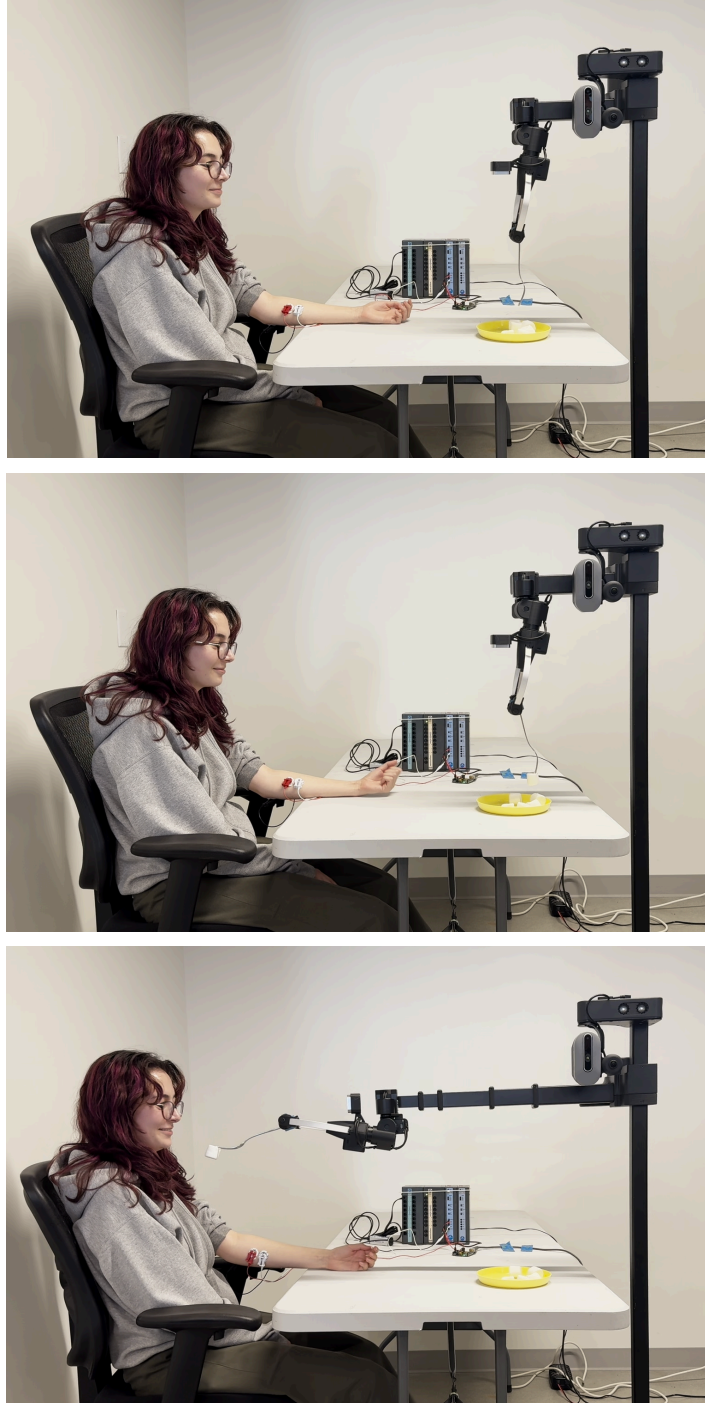


Figure 13. Three primary poses detailing the full robotic trajectory. Each activation signal from muscle contraction triggers predefined movement between successive poses before waiting for a subsequent command upon arrival. Video:  FinalVideo.mp4

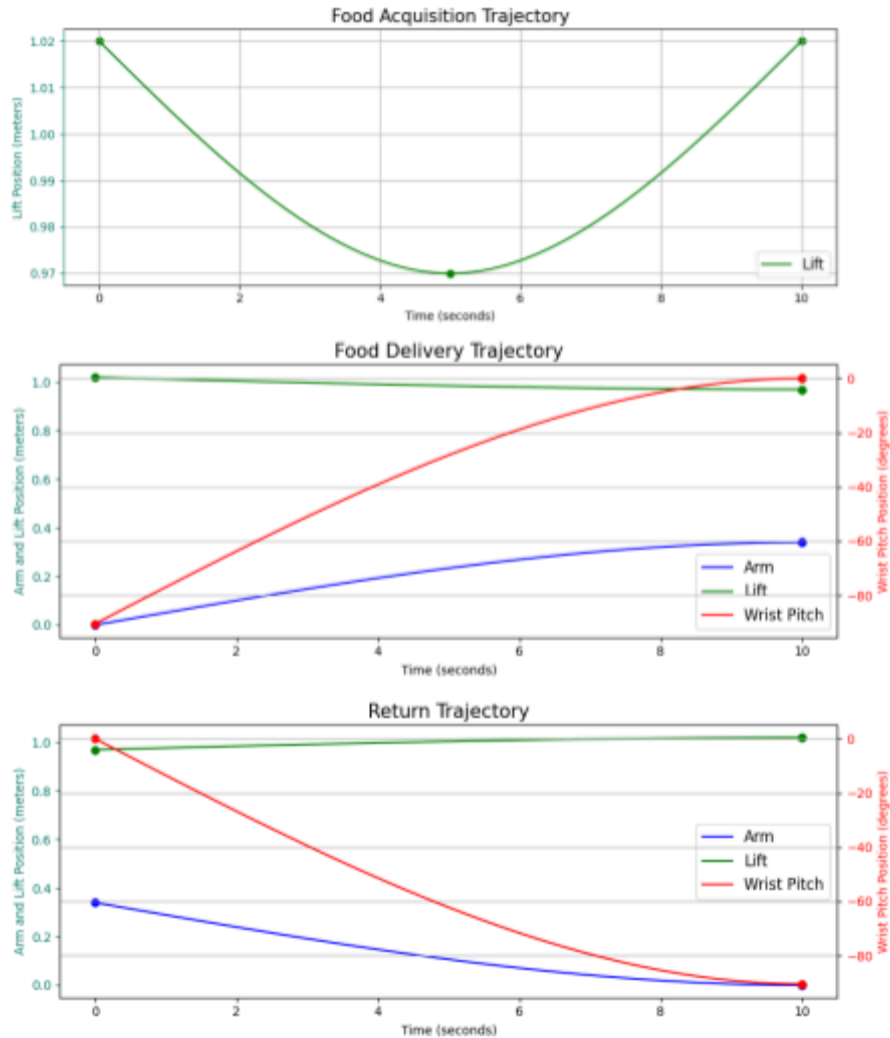


Figure 14. Full trajectory of all robotic joints in each of the feeding trajectories. The positions of each joint are plotted within the time it takes to perform the task. The wrist movement is in degrees, while the arm and lift are in cartesian space.

As seen in **Figure 14**, each trajectory for feeding was successfully designed from trajectories and incorporated into a “full body” trajectory, including multiple joints. Each of these trajectories was run on the stretch robot hardware and completed its tasks, as seen in the demo video and setup in **Figure 14**. Due to the cubic interpolation using the waypoints, a smooth and controlled trajectory was performed. All the joints move in unison and reach their desired endpoints at the same time, reducing any sporadic movement in the robot’s joints.

The trajectories also create a natural movement that is predictable and intuitive, as all the joints move to the user’s mouth in one motion: the lift reaching the height of the user, the arm extending to the user’s location, and the wrist orienting itself in a natural feeding position with the fork facing out towards the user. There were also no velocity or acceleration issues, as all the joints moved at slow enough speeds to reduce jerkiness or fast-moving joints, which could result in injuries and other safety issues.

Discussion

The results accomplished in this paper were completed under several assumptions. Robot trajectories were pre-defined, so the height and the distance of the user's mouth from the robot were known constants in our system. The design of the trajectories also assumed the food could be picked up with a fork, with a simple downward "stabbing" motion. Additionally, the design of our system assumes a few qualities about potential users. First, the user can either set up the EMG electrodes by themselves or with the assistance of a care partner. Additionally, users should be able to generate EMG signals with a superficial muscle.

While our setup is reliable and predictable within the specific configuration we outline, due to the constraints mentioned, there is a lack of adaptability to diverse real-world scenarios. Specifically, choosing different feeding locations around the house like a higher chair or food on a different table is currently not supported. The use of different foods that require different acquisition trajectories, like cereal or other granular materials would also be unfeasible.

EMG activation addresses the limitations of the current standard for accessible input methods, such as joysticks or touch control (Losey, et al. 2020). EMG can effectively capture user intent while being adaptable to the unique capabilities of each user, not requiring precise hand movement. It can be initialized and placed on any superficial muscle, preventing fatigue but encouraging movement and autonomy.

Despite the constraints, the core functionality exists and can be easily built upon. Using the built-in LiDAR and infrared sensors the Stretch 3 Robot has, the system could dynamically adjust to varying table heights or user positions by using computer vision to create adaptable trajectories. Adding a more versatile end effector for different foods, such as using a spoon or spork, while creating more trajectories for food acquisition, like scooping, slicing, and other motions can be programmed.

Conclusion

In conclusion, we successfully developed a proof-of-concept robotic system for assistive feeding with personalized EMG-controlled manipulation to acquire food and serve the operator, supporting varied levels and locations of muscular activation capability. This has laid the groundwork for a Stretch robot control program to be used and developed by any user to help bring greater independence for those with upper limb mobility impairments. Through the use of EMG as an input system, we have addressed several limitations of existing input methods, enabling more inclusive control. By allowing control through varied levels and locations of muscular activation, our system applies to a wide range of users with non-traditional movement abilities. We have expanded upon the functionality of the Stretch 3, by adding an assistive feeding feature to its already existing assistive tasks, thus creating a more complete assistive robot. Furthermore, because our program is open-source, we are allowing future scientists and researchers to build upon our work, helping to advance the existing research on assistive feeding. Assistive feeding robotics have a promising future as part of the movement to harness technology to create a more accessible and inclusive world.

Future Directions

Looking to the future, there are three main areas we would like to improve upon. Firstly, regarding hardware, we would like to integrate a second EMG input channel for more precise control. This second input will correspond to the extensor muscle coupling with the already measured flexor muscle, to enable repetition of the current step in the feeding cycle should it be desired. Further, we will refine the end effector to enable different utensils depending on the food type and expand the range of foods the user can eat using our system. Additionally, as user safety is always of the utmost importance, we will add an accessible emergency stop function that the user can easily activate at any time to pause the system.

Secondly, on the software side, developing a customizable, user-friendly GUI will allow users to adjust and personalize settings such as speed of motion and predefined trajectories for foods with various sizes and textures. Further, improving EMG signal processing will allow for a more natural and real-time robot interaction with the user. Lastly, we would like to conduct feeding trials on diverse participants representing a wide range of motor impairments to gather real-world feedback to improve our system. These adjustments will help us achieve our goal of advancing the field of assistive robotics whilst ensuring our system remains broadly accessible.

Appendix

I. EMG Setup Protocol

1. Identify a muscle that the user can flex for at least one second comfortably.
2. Exfoliate the skin with a towel or chemical treatment. Optionally remove hair from the skin area. Clean the skin over the desired muscle area with an alcohol wipe.
3. Plug an OUTISO cable into the top left port of the HLT100C labeled "1."
4. To connect the EMG electrodes to the amplifier, first plug in the singular black electrode into the center port labeled "GROUND."
5. Plug the red side of the red electrode wire into the VIN+ port and the black side into the upper SHIELD port.
6. Plug the white side of the white electrode wire into the VIN- port and the black side into the lower SHIELD port.
7. Gather the wires and set them to the side to keep them organized.
8. Peel sticker papers off of the pre-gelled electrodes.
9. Place the positive red electrode on the skin.
10. Flex the muscle and identify a spot 2 cm from the center of the red electrode along the same muscle. Place the center of the white electrode here.
11. Place the black electrode nearby on a bony area with little muscle, such as the elbow.
12. Ensure the amplifier is plugged in, and the green light is on to indicate that the power is on.
13. Wire OUTISO pins to Arduino Uno analog 0 and analog 1 pins.
14. Connect Arduino to computer using a USB cable and upload EMG Data Digital Filter Arduino Program (Appendix III).

II. Robot Setup Protocol

1. Turn on the Stretch robot and plug in the mouse, keyboard, monitor, and Arduino to the computer.
2. In terminal, run *stretch_free_robot_process.py*

3. Then run *stretch_robot_home.py* . Remember to remove clip during this step
4. In \autobite\src, run *main_trajectory.py* (before running, enter the correct Arduino port)
5. Place fork tool in the gripper, and press enter when ready to grip
6. Follow basic GUI commands to perform either individual trajectories, test Arduino input, or begin EMG + Trajectory movements
7. If in EMG mode, relax until the terminal displays that initialization is complete, then flex to begin feeding.

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